10.4. Formation of Differential Equations

10.4.1 Formation of Differential equations from Physical Situations

Now, we provide some models to describe how the differential equations arise as models of real life problems.

Model 1: (Newton's Law)



According to Newton's second law of motion, the instantaneous acceleration a of an object with constant mass m is related to the force F acting on the object by the equation $F = ma$. In the case of a free fall, an object is released from a height $h(t)$ above the ground level. Then, the Newton's second law is described by the differential equation $m\frac{d^2h}{dt^2} = f\left(t, h(t), \frac{dh}{dt}\right)$, where m is the mass of the object, h is the height above the ground level. This is the second order differential equation of the unknown height as a function of time.

Model 2: (Population Growth Model)



The population will increase whenever the offspring increase. For instance, let us take rabbits as our population. More number of rabbits yield more number of baby rabbits. As time increases the population of rabbits increases. If the rate of growth of biomass $N(t)$ of the population at time t is proportional to the biomass of the population, then the differential equation governing the population is given by $\frac{dN}{dt} = rN$, where $r > 0$ is the growth rate.

Model 3: (Logistic Growth Model)

The rate at which a disease is spread (*i.e.,* the rate of increase of the number N of people infected) in a fixed population L is proportional to the product of the number of people infected and the number of people not yet infected:

$$\frac{dN}{dL} = kN(L-N),\ k > 0.$$

EXERCISE 10.2

1. Express each of the following physical statements in the form of differential equation.
   (i) Radium decays at a rate proportional to the amount Q present.
   (ii) The population P of a city increases at a rate proportional to the product of population and to the difference between 5,00,000 and the population.
   (iii) For a certain substance, the rate of change of vapor pressure P with respect to temperature T is proportional to the vapor pressure and inversely proportional to the square of the temperature.

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